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RESEARCH ARTICLE

Effect of Molybdenum Disulfide (MoS_2) Reinforcement on the Structural, Thermal, Mechanical, and Tribological Analysis of Novel Polyamide 6,6 (PA 6,6) Composites

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ABSTRACT

Developing nanomaterial-based polymer composites by reinforcing fillers at various concentrations is an effective approach to achieving enhanced mechanical, tribological, and other desirable properties. In this study, novel MoS_2-reinforced PA-6,6 (PM) composites were fabricated using melt-mixing and injection molding techniques, incorporating MoS_2 fillers at different loadings. The surface morphology and microstructure of dispersed MoS_2 were analyzed using field-emission scanning electron microscopy (FESEM), while Fourier transform infrared spectroscopy (FTIR) confirmed the presence of key functional groups. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) results revealed the strong heterogeneous nucleation ability of MoS_2, significantly enhancing the crystallization temperature and thermal stability of the PM composites. Compared to pure PA-6,6, the melt-blended PM composites demonstrated improved hardness and tensile properties, with a ~21% increase in tensile strength observed for 5 wt.% MoS_2 filler. Detailed analysis of the fracture surface elucidated the fracture mechanisms, strengthening characteristics, and interfacial adhesion within the composites. Conversely, the impact resistance showed a decline relative to pure PA-6,6. Tribological performance, evaluated via linear reciprocating tribology (LRT) tests under varying abrading frequencies and normal loads, indicated improved friction and wear characteristics at 5 wt.% MoS_2. Worn surface analysis revealed a smoother morphology, confirming the composite's superior wear resistance followed by an adhesive wear mechanism.

1 | Introduction

The development and demand for micro/nanocomposites have interestingly increased due to the inherited unique and diverse properties under given conditions. Developing these composites by reinforcing one or more filler particles offers a means to desired characteristics tailored to specific requirements [1]. However, the selection criteria of such reinforcing materials for composite development involve a tough challenge for researchers that demands extensive data and credibility with authentic information from various sources on filler materials [2]. Polymeric matrices loaded with hard and nondissipative fillers can possess high stiffness and damping characteristics, which is an ideal structural properties. Over the last few decades, thermoplastic composites have interestingly increased due to their high-end applications with numerous marked mechanical and tribological applications such as seals, gears, bearings, and many automobile, electronics, and household components [3]. The lightweight characteristics of such composite materials have been proven to be better alternatives to metallic components [4, 5]. The tailoring properties of such polymeric materials with reinforcement of functional fillers have marked them as the most promising materials in the industrial arena. The study has revealed that polymer composites filled with only fiber contents can generally improve the mechanical properties [6]. However, filler-reinforced thermoplastic composites can effectively improve stiffness, decrease thermal expansion, and improve thermal properties and their long-term mechanical performance with reduced costs [7, 8]. Various materials, including glass fibers [9], carbon fibers [10, 11], glass beads, carbon black [12], wood flour [13, 14], etc., have been utilized as filler materials in thermoplastic composites [15, 16].

Furthermore, The mechanical, physical, and chemical properties of polymers are fundamentally governed by the structure and organization of their macromolecules [16, 17]. Consequently, a precise characterization of molecular order is essential for understanding the macroscopic characteristics of polymeric materials [18]. Polyamide-6,6 (PA-6,6) is a well-known semicrystalline polymer utilized under the engineering polymer section, which possesses a high melting temperature, high thermal stability, appreciable mechanical and tribological properties, along with high hardness characteristics as well as good dimensional stability [19]. The outstanding mechanical performance in terms of resistance and toughness has made PA-6,6 a potential candidate for its effective utilization in various industries [16]. However, its high moisture absorbance characteristics, low electrical conductivity, and high percentage of shrinkage limit its utilization in various aspects [16, 20]. Therefore, to overcome these shortcomings, the reinforcement of various fibers or filler materials in a certain weight percentage (wt.%) is essential to enhance their properties as per the required demand and application-based utilization [21, 22].

However, some of the most important and influencing inherited properties of filler materials are the thermal, mechanical, chemical, and physical properties of materials. Another important aspect that needs to be considered is the surface reactivity of the filler materials, which are necessarily required to be added to the solid/liquid phase in a definite quantity to carry out some desired functions [23, 24]. These filler materials offer enhanced lubrication benefits, facilitating smoother sliding relative movement between the mating parts. Machine parts in sliding or rolling contact conditions undergo wear over time due to friction. Therefore, there is a demand for improved materials possessing desired lubricating properties to extend machinery service life by reducing wear rate and friction coefficient [25, 26].

In this regard, several investigations have been carried out for the development of composite systems by reinforcing various filler materials into the polymeric matrix phase [27–29]. Among various 2D materials such as graphene, graphene nanoplatelets (GnPs), reduced graphene, extended graphene, graphite, and MoS_2. Out of them, MoS_2 is one of the most widely studied lubricating materials with fascinating properties [30]. MoS_2 is used as a solid lubricant material due to weak interlayer interactions, which permit the sliding of one layer over the other [31]. MoS_2 possesses a sandwiched structure containing layers of molybdenum sandwiched with two other layers of sulfur using a strong covalent bond between Mo and S and weak Van der Waals force of interactions among various layers [32]. Moreover, along with the luminescence and semiconducting properties, MoS_2 nanosheets have good potential to resist elastic deformation, as well as possess high elasticity and Young's modulus [33].

Despite its importance, only a few studies are available investigating the tribological performance of commercially available polymer composites filled with solid-lubricant fillers for industrial applications such as bearings, automobile parts, and hydropower applications [34–38]. Johnes and his coworkers [34] carried out an extensive study on friction and wear behavior under both dry and lubricated conditions and proposed a rating system for several commercially available bearing materials. Gawarkiewicz et al. [35] reported the friction coefficient and wear rates of four bearing materials under the simulating guide vane bearing operation. In both studies, the long-duration test was carried out, which lacks the surface analysis of transfer layer formation, and the possible wear and friction mechanism was not adequately discussed. Similarly, a study on the tribological performance of glass-filled polyamide 6 has also been carried out by Zaghloul et al. [39]. The literature studies the friction and wear behavior of PA6 and PA625GF composite under harsh abrasive environments and defines the optimal PV settings. The study reflects the superior antiwear characteristics of PA625GF composite compared to pure PA6.

The filler reinforcements in matrix material do not always reduce friction, but it also depends on the nature of the polymer matrix. However, these fillers are generally essential for achieving higher wear resistance, particularly against abrasion under high wear test conditions such as applied normal load (N), pressure (P), sliding speed (V), and frequency (f). The study of tribological behavior under these parameters is essential because it defines the durable operating range of polymer materials. Under this purview, the short carbon fibers, polyetheretherketone (PEEK), ceramic nanoparticles, and epoxy-reinforced polyphenylene sulfide (PPS) based hybrid composite exhibited lower friction and wear behavior with increasing PV values [40–42]. Similarly, a nano-short carbon fiber/epoxy/SiO_2 hybrid nanocomposite demonstrated a low coefficient of friction of 0.06 at 24 MPa.m/s [43]. In another study, Chang and coworkers reported that the lower CoF values decreased from 0.41 to 0.012 for PEEK hybrid nanocomposite filled with nano-short carbon fiber/graphite/TiO_2 filler contents with increasing PV values from 1 to 4 MPa.m/s [44]. Thus, exploring the ability of polymeric composite materials to attain ultra-low wear and friction is crucial. It is generally believed that a high-performing transfer layer film formation is directly related to minimizing the friction between the tribo pair. However, the development of transfer layer film on the counterface surface during the sliding process of the polymeric composite is a complex procedure driven by chemical and tribo physical variables [45–50]. Therefore, it is important to study the functionality and formation of the transfer layer tribo film. Also, the previous studies were mostly carried out for the pin-on-disk tribology test, and very few works on the pin-on-plate tribology test were reported. No such study on the linear reciprocating tribology (LRT) test using a ball-on-plate tribo pair arrangement under load and frequency test parameters was found in the previous literature.

Therefore, the present study aims to evaluate the structural, thermal, mechanical, and tribological properties of

PA-6,6-based thermoplastic composites reinforced with MoS_2 solid lubricant particles at varying weight percentages (1%, 3%, 5%, and 10 wt.%). The research investigates the efficiency of the developed PM composites compared to pure PA-6,6, with an emphasis on their tribological performance, including friction and wear behavior, under different conditions. Mechanical fracture behavior is assessed through fractured surface morphologies using FESEM micrographs, offering insights on tensile fracture phenomenon, rarely reported in the literature. The primary objective is to determine the optimal load and frequency parameters in the LRT test, utilizing a steel ball as the counterface, to achieve improved friction and wear performance of the PM composites under different load conditions and linear frequencies. The study aims to establish safe operational boundaries for the PM composites and demonstrate the effects of MoS_2 reinforcement on tribological behavior, such as transfer layer formation and tribochemical reactions. The combination of these parameters for each composition of the composite material provides a guideline that exceeding these limits increases friction between the mating surface, leading to accelerated wear, thermal degradation, and localized melting. Due to this phenomenon, the mechanical properties of the material get potentially compromised. This work presents, for the first time, a comprehensive analysis of these parameters to guide the use of PA-6,6 composites in various tribological applications, ensuring consistent performance.

2 | Materials and Methods

2.1 | Materials Used

In the present study, the commercially available Polyamide-6,6 granules (PA-66) (unfilled and natural color, PXR-01 NC) were procured from Celanese India Pvt. Ltd. (INDIA). Additionally, a solid lubricating agent, molybdenum disulfide (MoS_2), was sourced from Sigma-Aldrich Chemical Pvt. Ltd., INDIA.

2.2 | Compounding and Injection Molding

Prior to the compounding, the PA-6,6 polymer granules and fillers were kept for drying at 90°C for 4–6 h in a hot air-circulated oven (MCP, HEK, Germany). Subsequently, various batches of the MoS_2 filler material for different compositions were prepared by weighing and dry-mixing with the PA-6,6 matrix. This step minimized the risk of moisture-induced issues and enhanced the flow properties during processing, contributing to better dispersion. Further, the premixed batches were processed in a Haake mini-lab co-rotating twin screw extruder (Thermo Fisher, Germany), well known for its high shear capabilities. The extrusion process was conducted at a temperature range between 275°C and 285°C, ensuring the easy extrusion of the polymers while avoiding thermal degradation. Additional care was taken for its controlled screw speed of 125–130 rpm to generate sufficient shear forces to break the MoS2 filler agglomerates and evenly distribute the filler particles throughout the polymer matrix. For improved composite behavior, the material was processed for 2–3 min to allow adequate mixing time without compromising the thermal stability of the materials. After extrusion, the extruded melt-blend was collected in a cylinder maintained at 275°C–300°C [15, 16] and injected into a Haake mini-jet injection molding machine at a high pressure of 600 bar for 10 s of injection time. This ensured the MoS_2 dispersion achieved during extrusion was retained in the final composite specimens. The effectiveness of the dispersion was evaluated using FESEM images, as delineated in Figure 2b, which confirm that the MoS_2 particles were well dispersed within the matrix, with minimal agglomeration observed. This process aims to fabricate the PM composite samples by reinforcing a different weight percentage (wt.%) of MoS_2 filler particles suitable for various testing procedures. This method effectively combines mechanical dry mixing, melt processing with high shear, and precise control of processing parameters to achieve a uniform dispersion of MoS_2 in the PA-6,6 matrix. The systematic approach ensures that the composite exhibits consistent properties and reliable performance.

3 | Characterization Techniques

3.1 | Density and Microhardness Measurement

The density of the prepared composite samples was evaluated by determining weight over volume. The weight of the composite samples were obtained by an electric weighing balance (Mettler Toledo, Germany) with an accuracy of 0.0001 mg. The volume of the solid polymer sample was determined by measuring the volume of water displaced when the sample was completely immersed in water by pushing it in using the tip of a needle in a graduated burette. The error probability in the volume and density measurements was calculated to be 5%. Microhardness testing was conducted using a microhardness tester with an indentation force of 150 gf (980.66 mN) applied for 15 s of dwell time. Five indentations were made randomly on each sample, and the average was taken as the hardness value (in Vickers Hardness units). For each measurement, three different samples were tested, and the reported data represent the averages of these three samples.

3.2 | Surface and Microstructural Characterization

The microstructures of the developed composites and the surface morphologies of tensile fractured samples were analyzed using the Field Emission Scanning Electron Microscope (FE-SEM) (Sigma 300, Carl Zeiss Microscope Ltd., Germany) in high vacuum at the voltage of 10–20 kV. The elemental compositional analysis of the prepared composite samples was conducted using an additional energy-dispersive spectroscopy (EDS) attachment connected to the FE-SEM facility. The EDS analysis was performed using the EDAX (Ametek) system. The samples were carried out for preliminary metallization with a gold coating using Quorum (Q150R ES), UK, to ensure proper conductivity of the observed areas.

Moreover, the worn surface morphologies were obtained from the worn surface of the samples posttribology and examined using the scanning electron microscopy JEOL JSM-6390LV (SEM), Japan. Before SEM micrographs, the metallization process of the worn surface was carried out with a Platinum coating using JEOL JFC-1600 auto fine coater.

3.3 | Structural Characterization

Further, the identification of the presence of different functional groups in the developed PM composites, as well as the chemical interaction of MoS_2 filler particles with the PA-6,6 matrix phase, was studied using the Fourier Transform Infrared Radiation (FTIR) spectral measurement technique at 32 scans with a resolution of $4\,cm^{-1}$ across the mid-infrared light band within the spectral wavenumber range of $4000–400\,cm^{-1}$ using the Perkin Elmer 100 spectrometer equipped with an ATR module.

The x-ray diffraction (XRD) patterns of pure PA-6,6, and its PM composites were acquired utilizing a Rigaku smart lab, a 9 kV x-ray diffractometer fitted with a Cu-Kα radiation ($\lambda = 1.54184$ nm) x-ray tube, and an end-window counter detector capable of detecting α, β, γ, and x-rays. Data were recorded over a 2θ range of 10°–80° with a scan step of $2\theta = 5°min^{-1}$.

3.4 | Thermal Characterization

The thermal properties of the prepared composites were assessed using differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA), employing a Discovery DSC 25 (Waters, USA) and a DTG-60 (Shimadzu Corporation, Japan) TGA/DTA analyzer, respectively. In DSC analysis, the Heat–Cool–Heat method was adopted. Initially, the specimens underwent heating up to 300°C and were held at isothermal condition for 1 min to eliminate any prior thermal effects. Subsequently, the samples were cooled to ambient room temperature (RT). Both the heating and cooling processes were conducted under a nitrogen (N_2) atmosphere, with a heating/cooling rate of 10°C/min. In the present work, the DSC analysis focused on the second heating thermogram and the obtained data on the melting (T_m) and crystallization (T_c) temperatures, melt enthalpy or enthalpy of fusion (ΔH_f) have been tabulated in the subsequent section, and the degree of crystallinity (% X_c) from the DSC thermograms was calculated using Equation (1) as follows.

$$X_c(\%) = \left(\frac{\Delta H_f}{\Delta H_m^0}\right) W_f \times 100 \tag{1}$$

Where, $\Delta H_m^0 = 190\,J.g^{-1}$ for 100% crystalline PA-6,6 [51], and W_f indicates the wt.% of the PA-6,6 in the composite.

Additionally, the TGA analysis was carried out to assess the thermal stability of the developed PM composite. The experimental investigation was conducted using a platinum pan containing samples (around 3–5 mg) in an inert N_2 atmosphere with an N_2 flow rate of 50 mL/min. The samples underwent heating from RT to 800°C at a heating rate of 10°C/min, and simultaneously, the alterations in their weight resulting from the decomposition of the sample were continually observed.

3.5 | Mechanical Property

To evaluate the tensile properties of the developed composite, a tensile test was conducted following ASTM D638 guidelines. The samples were tested by employing an Instron-3366 universal testing machine (UTM) equipped with a dynamometer-based load cell capable of handling loads up to 30 kN. The cross-head speed during the tensile test was 5 mm/min. Furthermore, to facilitate impact testing, the Izod impact test was conducted at room temperature utilizing an Izod impact tester IT504 (Tinius Olsen, INDIA) equipped with an 11.224 J hammer, following ASTM D256 standards. The impact test results of the developed composite were analyzed to assess their impact strength under the V-notched (at a 45° angle to a depth of 25 mm) opening mode (tension mode). It is important to note that five samples were considered to undergo testing for each test, and the reported values represent the average results.

3.6 | Tribological Property

To assess the tribological performance of the PM composites, the wear test was carried out using a multitribometer TR-705 (DUCOM, USA) equipped with the ball-on-plate tribo pair arrangement. Wear scars were generated by the repeatedly sliding action of a steel ball, with 6 mm dia. (AISI-E-52100 (100Cr6)), counterface surface against the polymer composite plate. All tests were performed under indoor laboratory conditions at 20°C–25°C, with the relative humidity maintained at 50%–60%. All samples were tested for minimum (5 Hz) and maximum frequency (25 Hz) under three different normal load conditions of 10, 30, and 50 N. Further, a high precision weighing balance with an accuracy of ±0.0005 g was utilized to weigh the test samples before and after the test to determine the weight loss. Figure 1 illustrates a schematic representation of the ball-on-plate tribo pair employed in the testing setup. These experiments aimed to assess the coefficient of friction (CoF) and specific wear rate (W_{SR}) for composites containing various wt.% of reinforced MoS_2 filler contents. Following CoF evaluation, the specific wear rate for the prepared composite samples was calculated using the relation provided in Equation (2).

$$W_{SR} = \left(\frac{V}{P \times D}\right) \tag{2}$$

where W_{SR} shows the specific wear rate in mm^3/N-m, V is the wear volume in mm^3, P denotes the applied normal load in N, and D is the sliding distance in mm.

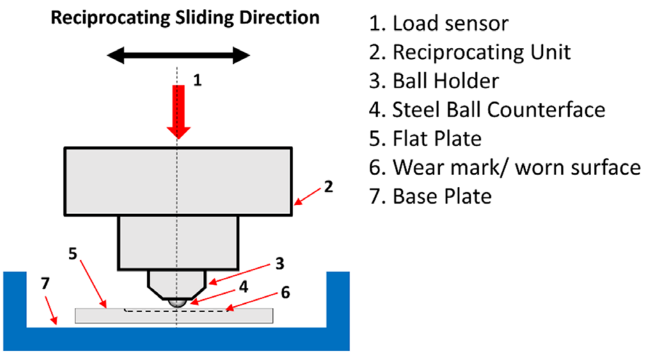

FIGURE 1 | Schematic diagram of computerized ball-on-plate tribo pair arrangement. [Color figure can be viewed at wileyonlinelibrary.com]

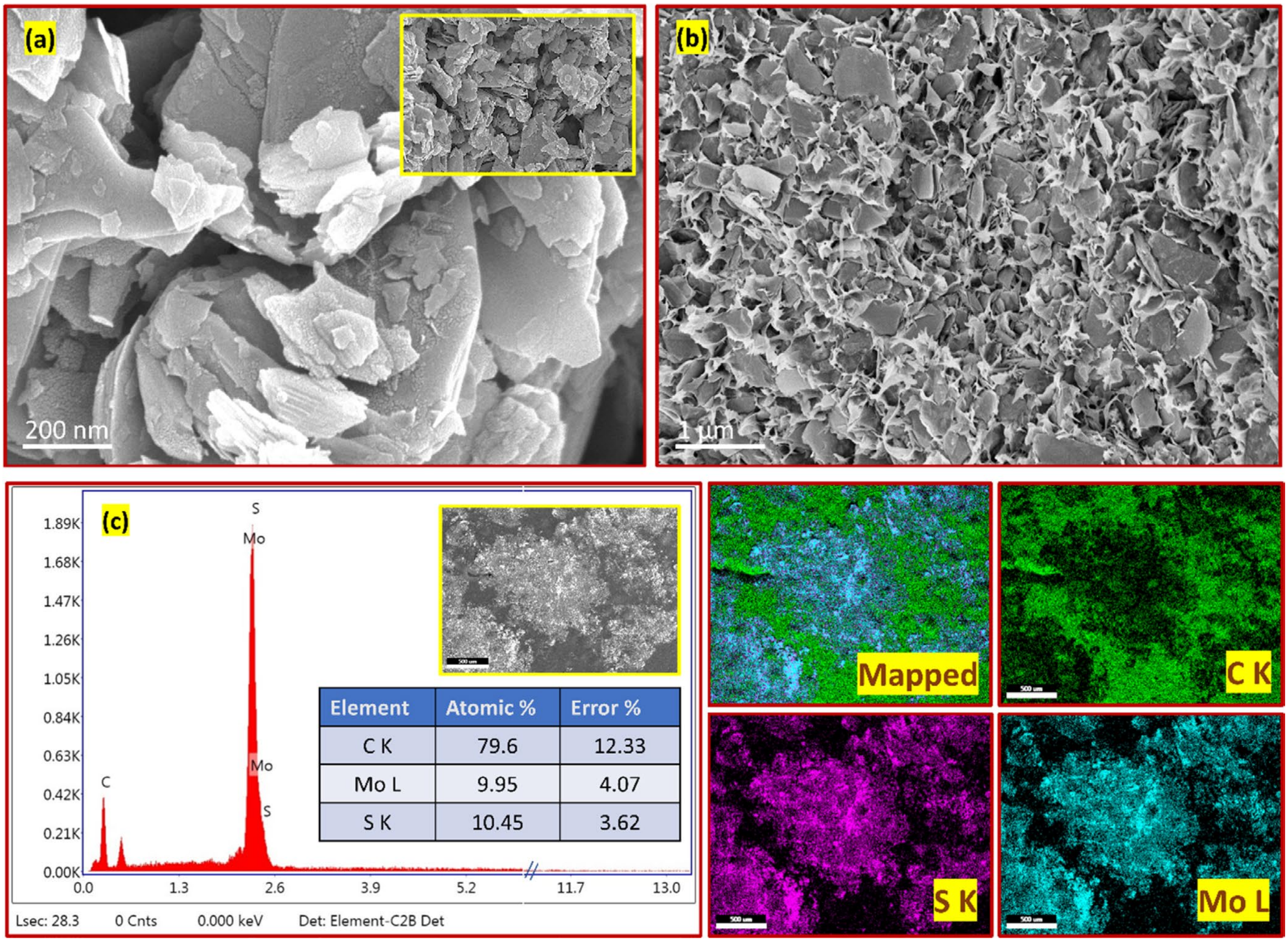

FIGURE 2 | FESEM images illustrate (a) the morphology of MoS_2 powder, (b) the dispersion of MoS_2 in the polymer matrix, and (c) the EDX element mapping of the prepared composite. [Color figure can be viewed at wileyonlinelibrary.com]

3.7 | Wettability Test Analysis

Furthermore, the wettability tests were conducted using a sessile drop technique using a Dataphysics, Germany, optical contact angle measurement machine, which captures images of water droplets on various locations on sample surfaces with a camera. All the samples were tested for water contact angle (WCA) at room temperature in five different locations on the finished sample surface, and the mean of all the WCA values was reported here. During the WCA test, the attached syringe forms a liquid (water) droplet of 2 μL and drops the droplet delicately onto the sample surface. A charge-coupled device (CCD) camera was utilized to capture the image of the dropped droplet. These images were then analyzed using OCAH 230 image processing software. An image was captured immediately to minimize errors caused by water evaporation.

4 | Result and Discussions

4.1 | Morphological Observations

Both pure PA-6,6 and MoS_2 powder materials were processed by the melt-mixing technique to fabricate the polymer composite samples. Figure 1 illustrates the FESEM images of the MoS_2 and the morphological observations of the PM composite. The morphological observations of the MoS_2 (See Figure 2a) clarify a layer-wise sandwiched structure, ensuring the 2-dimensional geometry with a stacked layers arrangement. Additionally, the incorporation of MoS_2 filler particles in the PA-6,6 matrix phase exhibits the uniform dispersion of MoS_2 filler particles over the entire region of the PA-6,6 matrix without agglomeration, indicating the proper mixing attributing to an effective filler–matrix interaction, as shown in Figure 2b. However, it is well known that achieving good interfacial dispersion of the filler particles over the entire volume of the matrix phase is crucial for the successful preparation of composite samples that are responsible for further enhancements of the composite's properties [52]. Furthermore, the EDS analysis, along with the elemental mapping, confirms the presence of MoS_2 filler contents in the polymer composite and concludes the uniform elemental distribution of the filler particles.

4.2 | FTIR Analysis

Fourier transform infrared (FTIR) spectroscopy is an effective method for identifying the presence of various functional groups of the prepared samples, differences in their chemical structures,

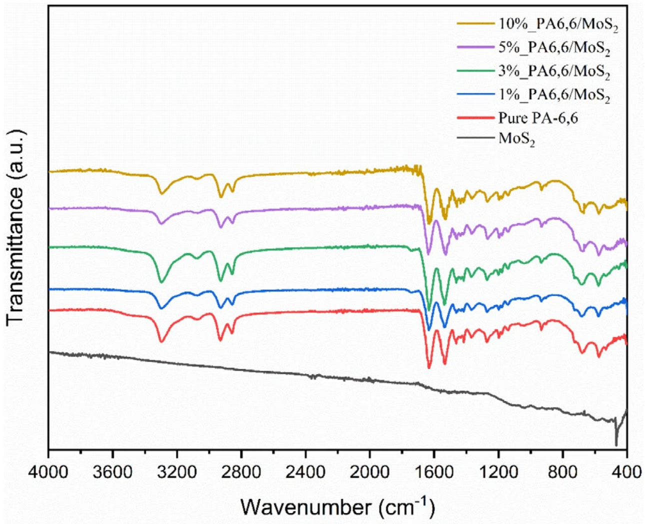

FIGURE 3 | FTIR spectral analysis of PA-6,6/MoS_2 composite indicating infrared peaks. [Color figure can be viewed at wileyonlinelibrary.com]

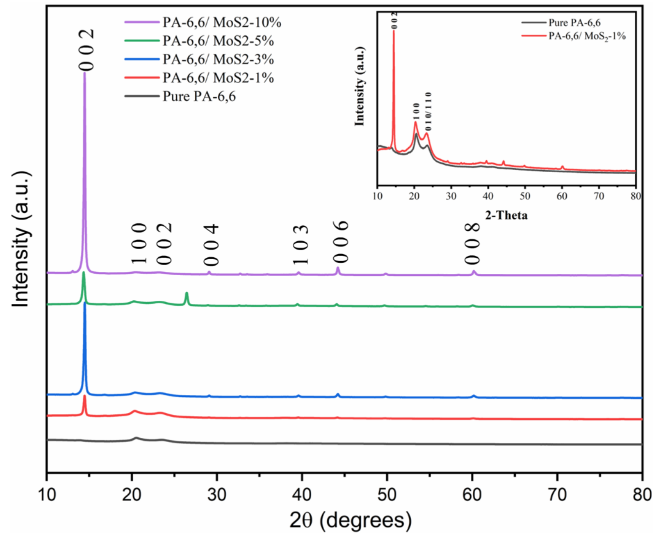

FIGURE 4 | XRD pattern of MoS_2 and its PM composite with different concentrations of MoS_2 in the PA-6,6 matrix. [Color figure can be viewed at wileyonlinelibrary.com]

and assessing the dispersion of filler particles in polymeric systems, serving as a supplementary technique to XRD [53, 54]. Figure 3 depicts the FTIR spectral analysis of the pure PA-6,6 and its PM composites. Figure 3 exhibits an almost similar FTIR spectral pattern of pure PA-6,6 and its PM composites. The spectral range observed around 3295–3316 cm^{-1} was found present in pure PA-6,6 and all MoS_2-filled PA-6,6 composites due to —NH stretching from amino groups in PA-6,6 [16, 55]. The medium peak observed at 2927 cm^{-1} is due to the —CH_2 asymmetric methyl group, and again, the spectral band observed at 2857 cm^{-1} in PA-6,6 and all its composites for —CH_2 symmetric stretching of the methyl group in PA-6,6 [56].

Additionally, the PA-6,6 and all its PM composites were observed to have two strong peaks detected at the spectral range of 1540–1650 cm^{-1} [54, 57]. The spectral peak at 1636.65 cm^{-1}, C=O stretching from the carbonyl group, can bind with the amino group to form intramolecular hydrogen bonding. This interaction causes the C=O stretching, typically between the spectral range 1760 and 1665 cm^{-1}, to shift to 1636.78 cm^{-1} [58]. Meanwhile, the amide II at spectral band 1533.58 cm^{-1} appears due to C–N stretching and N–H bonding. The MoS_2 spectrum shown in Figure 2 shows two peaks at very high wavelengths that are representative of medium O–H due to the presence of water in the structure of MoS_2: absorbance peaks of S=O and Mo–O were observed at spectra 1124.94 and 676.19 cm^{-1}, respectively [59].

4.3 | XRD Analysis

The XRD analysis was further conducted for the PA-6,6 and its PM composites to confirm the intercalation effect of MoS_2 with the PA-6,6 matrix. The amorphous phase of pure PA-6,6 polymer matrix was identified at 20.4° and 23.593° with crystal plane position at (1 0 0) and (0 1 0/1 1 0) in the XRD pattern, as shown in the insert image provided in Figure 4, suggesting the presence of a very small crystal [60]. Additionally, due to MoS_2 filler contents, the broader peak presented in PA-6,6 at a lower 2θ value of 14.01° was found to be shifted to 14.37° with increased peak intensity, exhibiting the presence of a layered structure MoS_2 in the PA-6,6 matrix phase. Additionally, the characteristic diffraction angle with the most prominent peak was observed to be at 14.37° (interlayer spacing of ~6.155) with a crystal plane position at (0 0 2) [61]. At lower wt.% of MoS_2 introduction, the peak was found to disappear, but other characteristic peaks were formed at higher loading of MoS_2 contents. Other diffraction peaks of the PM composites were identified at the 2θ values of 29.12° (d = 3.0737), 39.52° (d = 2.277 Å), 44.15° (d = 2.049 Å), and 60.14° (d = 1.537 Å). The decrease in interlayer spacing is attributed to the more compact structure formation of MoS_2 during its processing with PA-6,6.

This common effect is identified due to strong interfacial interaction between the polymer and MoS_2 sheets. Additionally, during the composite preparation, MoS_2 is often exfoliated to increase its surface area and improve its dispersion within the polymer matrix phase. Subsequently, the restacking of its layer within the PA-6,6 matrix takes place, leading to a decreased interlayer spacing as the layers become densely packed. Compared to the exfoliation of MoS_2 reported in the literature, the characteristics peak identified at 14.3° is difficult to disappear, mainly due to the re-stacking of MoS_2 layers during the water/solvent removing process. This study greatly emphasized avoiding the aggregation of MoS_2 filler materials due to its reinforcement in long-chain PA-6,6 polymer matrix. The present findings were found to be consistent with the previous literature [62–64]. Further, with the increased content of MoS_2, the intensity of the diffraction peak becomes stronger, so it can be concluded that the addition of MoS_2 helped to improve the crystallinity of PM composites.

4.4 | Thermogravimetry Analysis

The TGA thermogram of PA-6,6 and its PM composites, as shown in Figure 4, and the corresponding thermal stability data

TABLE 1 | Thermal data obtained from the TGA analysis for PA-6,6 and its PM hybrid composites.

Sample code	$T_{10\%}$ (°C)	$T_{50\%}$ (°C)	T_{max} (°C)
Pure PA-66	357.6400	450.0700	566.7700
PA-6,6/MoS_2_1%	408.3800	453.9000	569.4100
PA-6,6/MoS_2_3%	403.8300	450.3500	583.3200
PA-6,6/MoS_2_5%	402.5400	454.9000	590.0600
PA-6,6/MoS_2_10%	392.7400	451.2900	595.3800

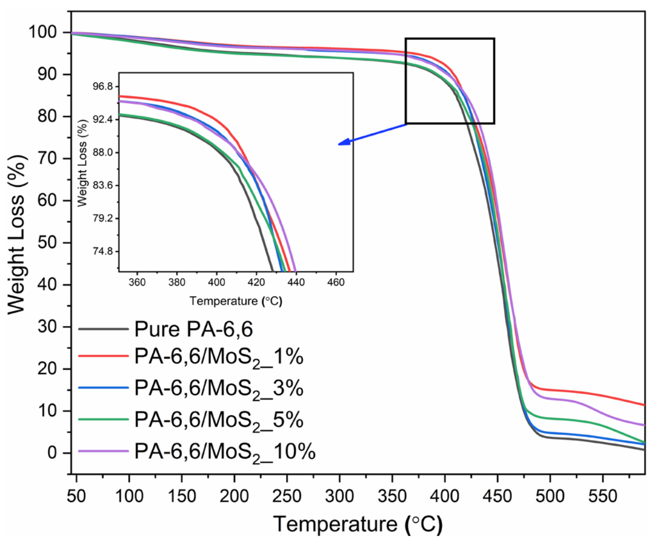

FIGURE 5 | TGA curve of PA-6,6 and its PM composites under N_2 environment. [Color figure can be viewed at wileyonlinelibrary.com]

are tabulated in Table 1. Thermogravimetric analysis (TGA) was conducted to evaluate the thermal behavior of the PA-6,6 polymer matrix reinforced with MoS_2 filler particles at varying wt.%. The thermal stability of a polymer composite system is primarily influenced by its morphological characteristics, the crystallinity of the matrix phase, the size of the fillers loading, the interfacial adhesion between the fillers and the matrix, and the degree of filler dispersion throughout the matrix [65–67]. From Figure 5, a single-step thermal decomposition can be seen, indicating a significant improvement in the thermal stability of the PM composites on MoS_2 filler addition at various concentrations. The thermal breakdown of the PA-6,6 and its PM composites was found to be in the range of ~450°C–455°C for 50% weight loss (T_{50}). The MoS_2 filler loadings at various wt.% (1, 3, 5, and 10 wt.%) were found to have a significant enhancement in wt. loss (%) at T_{10}, T_{50}, and T_{max} thermal decomposition. The thermal breakdown with 10% weight loss (T_{10}) for PA-6,6 was identified at 357.64°C, while enhancements were observed in the reinforcement sequence of MoS_2 in PM composites. The main degradation was identified in the temperature range of ~355°C–500°C.

Similar increasing behavior in thermal stability was also seen at T_{max}, at which the mass loss is 99%, and the main degradation temperature was found to be in the range of ~550.77°C–600°C, which shows an increase of 2.64°C, 16.55°C, 23.29°C, and 28.61°C compared to pure PA-6,6 polymer matrix on successive reinforcements of 1 wt.%, 3 wt.%, 5 wt.%, and 10 wt.%, respectively. The addition of 1 wt.% MoS_2 does not significantly influence the thermal degradation behavior of PA-6,6. However, as the MoS_2 filler content increases, the thermal degradation resistance of the composite system improves progressively. This enhancement can be attributed to the barrier effect of well-dispersed two-dimensional MoS_2 particles, which restrict the thermal movement of polymer chains, slow down the mass loss of the PM composite, and inhibit the release of volatile degradation products [68]. This improved thermal stability often correlates with enhanced mechanical properties such as higher tensile strength and modulus and exhibits better wear resistance characteristics in tribological applications [16, 69, 70]. The enhanced wear performance of thermally stable composites can be attributed to their reduced tendency to degrade or soften under frictional heat generated during sliding against the counterface surface. This stability promotes efficient heat dissipation during the wear process, maintaining a consistent friction coefficient and wear rate even at elevated temperatures. Consequently, the occurrence of severe adhesive wear is significantly minimized [71, 72]. Similarly, studies with improved thermal behavior have previously been reported that support the current findings. Chhetri and her coworkers [73] reported improved thermal stability in linear low-density polyethylene (LLDP) composites due to MoS_2 reinforcement. Another work carried out by Zhou and his coworkers [74] reported thermally stable behavior of the modified MoS_2/PS composite with admirable reinforcing ability. The study demonstrated that at 3 wt.% of CTAB-MoS_2 addition, the $T_{50\%}$ of the sample is increased by 60°C. Such improved thermal behavior indicates the barrier effect and better compatibility of the polymer matrix with organic or inorganic filler materials, forming a strong interfacial interaction and good interfacial adhesion between the filler and matrix phase [30].

4.5 | Differential Scanning Calorimetry (DSC)

In addition to XRD, DSC was also used to analyze the crystallinity of the polymer. The melting enthalpy of the polymer is proportional to its crystallinity. High crystallinity corresponded to the increased melting enthalpy [75]. Figure 6 demonstrates the DSC thermograms of the 2nd heating and the cooling curves, and the thermal data obtained from the DSC analysis are tabulated in Table 2. Pure PA-6,6 exhibited a melting temperature (T_m) peak at 262.24°C during the first heating cycle, and a crystallization temperature peak (T_c) was observed at 238.74°C. The Pure PA-6,6 polymer matrix is a semicrystalline polymer material, with a relative degree of crystallinity (X_c %) found to be ~28%. Furthermore, a decrease in the enthalpy of crystallinity and an increase in the melting enthalpy of the PM composites was observed, which is due to the reinforcement of MoS_2 filler particles, which destroys the crystallinity of PA-6,6 and interacts with its molecular chains to form a compatible structure during the preparation process. As a result, enhanced T_m values of the composite were observed [76]. Additionally, a second heating cycle, as shown in Figure 6a, exhibited the two crystal forms of PA-6,6: the α-crystalline regions characterized by hydrogen bonds between antiparallel chains and γ-crystalline regions characterized by hydrogen bonds between parallel chains, for which the corresponding T_m peaks were observed at 252.82°C

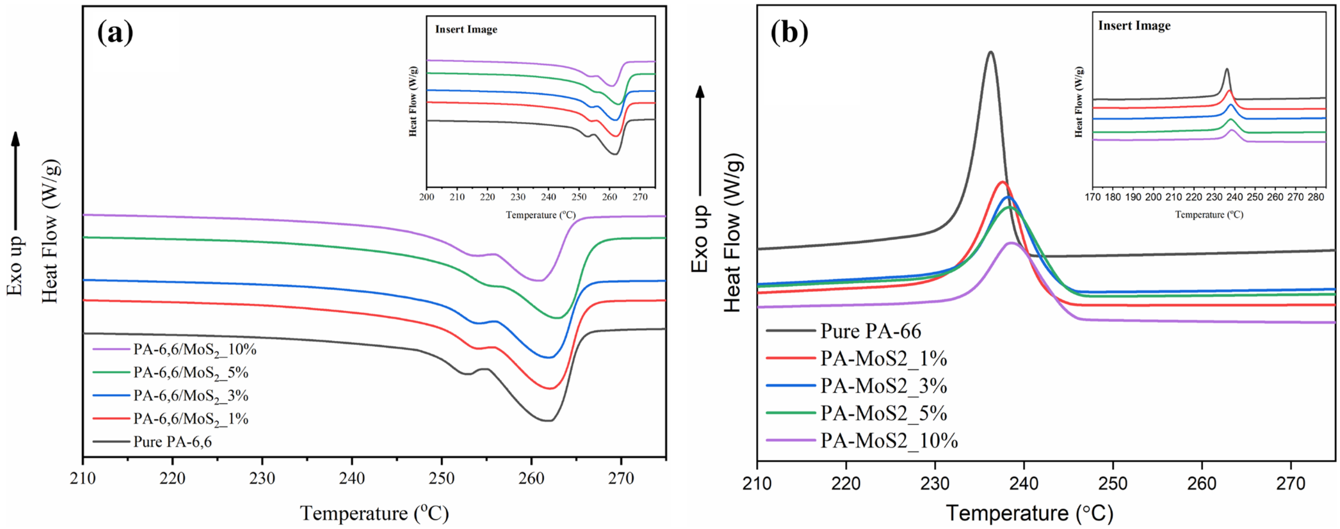

FIGURE 6 | DSC thermograms indicate (a) melting during the second heating cycle and (b) crystallization during the cooling cycle of PA-6,6 and PA-6,6/MoS_2 composite. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 | The DSC thermogram scans data for the 2nd heating and cooling cycles of PA-6,6 and its PM composites.

Sample code	T_m	ΔH_m	T_c	ΔH_c	X_c (%)
Pure PA-6,6	262.240	62.703	238.740	47.515	27.745
PA-6,6/ MoS_2-1%	262.990	66.752	241.730	43.765	29.023
PA-6,6/ MoS_2-3%	263.080	66.632	243.200	38.944	28.970
PA-6,6/ MoS_2-5%	264.410	54.881	244.470	40.473	29.992
PA-6,6/ MoS_2-10%	261.660	56.029	243.200	31.138	24.360

and 261.78°C [77, 78]. It is worth noting here that the MoS_2 reinforcement in PA-6,6 imparts a positive influence on the melting temperature and crystalline nature of the PM composite that also promotes the formation of the γ-crystal in the PA-66 matrix, and the proportion of γ-crystal also increases by increasing the MoS_2 filler contents in the matrix phase. In the present findings, the percentage crystallinity of the PM composite was found to be increased by approximately 8%, mostly due to the MoS_2 addition. This further provides a better insight into the enhanced mechanical properties such as mechanical strength, stiffness, and toughness. At higher wt.% of MoS_2 addition, a decrease in crystallinity was identified. This may be due to the depressed crystallization behavior of the PA-6,6 polymeric chains [77]. Furthermore, at higher concentrations, the reinforcement of MoS_2 can interfere with the hydrogen bond within the polymer, thereby reducing its crystallization capacity [79]. While MoS_2 particles act as nucleating agents at lower concentrations, excessive amounts can interfere with the molecular alignment, decreasing overall crystallinity. This reduction is attributed to the steric hindrance imposed by MoS_2, which restricts the mobility of polymer chains during the crystallization process. Additionally, the interaction between MoS_2 fillers and the PA-6,6 matrix at the interfacial level can hinder the development of crystalline regions. This interaction weakens stress transfer and bonding at the interface, resulting in the formation of less ordered structures and a consequent decline in crystallinity [80].

Furthermore, Figure 6b indicates the cooling cycle of the PM composite. The study of the cooling thermograms depicts an increase in crystallization temperature. This enhancement in the crystallization temperature of these PM composites may be attributed to the nucleating effect of the MoS_2 filler. The presence of MoS_2 particles behaves as a heterogeneous nucleating site, which facilitates the crystallization process of the polyamide matrix at a higher temperature than pure PA-6,6. This enhancement in nucleation leads to a higher degree of crystallinity and improved thermal stability, as MoS_2 reduces the energy barrier for crystal formation, thereby promoting earlier and faster crystallization. From previous research, it has been shown that fillers like MoS_2 provide more nucleation sites within the polymer matrix, leading to an increase in crystallization temperature and a reduction in overall crystallization time. This phenomenon is supported by studies where the incorporation of MoS_2 in polyamides improved crystallization behavior due to the enhanced nucleation efficiency of the filler particles [61, 81].

4.6 | Mechanical Properties

4.6.1 | Density and Hardness

Table 3 shows the measured densities of the pure PA-6,6 and PM composites prepared by adding different wt.% of MoS_2 concentrations. The densities of all nanocomposites were found to be higher than the pure PA-6,6 polymer matrix. The highest density of the PM composites was found to be 1.2407 g/cc at 10 wt.% of MoS_2 filler addition compared to all other compositions (1wt.%, 3wt.% and 5 wt.%) of nanocomposites. This is because the MoS_2 filler possesses a higher density

TABLE 3 | Density and hardness of PA-6,6 and its PM composites.

Sample code	Density (g/cc)	Hardness (Vickers, HV)
Pure PA-6,6	1.1373	18.2
PA-6,6/MoS_2-1%	1.1403	21.4
PA-6,6/MoS_2-3%	1.1620	24.1
PA-6,6/MoS_2-5%	1.1773	26.5
PA-6,6/MoS_2-10%	1.2407	29.3

of about 5.06 g/cc than pure PA-6,6 (1.14 g/cc). Additionally, the density of thermoplastic composites plays a vital role The density of thermoplastic composites plays a critical role in determining their mechanical, thermal, and tribological properties. The higher density implies a good number of reinforced filler particles, leading to improved mechanical strength, stiffness, and toughness. However, this also introduces a brittle nature in the material. Due to this, the maximum loading of the reinforcement further produces an opposite effect and reduced properties [82]. Furthermore, the higher density of the composites or the reinforcing filler materials enhances the thermal conductivity due to their higher thermal diffusivity [83], leading to improved heat dissipation with higher density [84]. Further, the higher thermal stability with improved heat dissipation characteristics produces a correlation with the reduced coefficient of thermal expansion, which restricts the mobility of the polymer chains due to the successively increased concentration of the filler particles and higher density [85, 86]. This also influences the friction and wear behavior of the polymer composites. The higher density of the composites demonstrates improved wear resistance characteristics and produces a reduced friction coefficient by altering the surface contact and energy dissipation mechanism [70]. For instance, the nylon composites filled with HBN enhance the friction and wear characteristics [54, 69, 87, 88]. Similarly, the denser composites with graphite, graphene, or molybdenum disulfide-reinforced polymers tend to exhibit a lower friction coefficient due to enhanced lubrication [70].

Furthermore, the Vickers microhardness tester was carried out to measure the hardness of the prepared PM composite samples. The average values recorded during the test are given in Table 3. The hardness of the PM composites was found to increase with the inclusion of MoS_2 filler particles in different wt.%. From Table 3, it was observed that the hardness of the prepared nanocomposites was greatly increased by increasing the filler contents from 1 wt.%–10 wt.%. At a higher MoS_2 concentration of 10 wt.%, the highest value of the microhardness was recorded to be a value of 29.3 HV. This increased hardness of these PM composites can be attributed to the influence of MoS_2 addition in the PA-6,6 matrix. The filler particles act as barriers to localized deformation, thereby improving the hardness of the composite samples with enhanced resistance to indentation. The rigid structure of MoS_2 facilitates uniform distribution of the applied load, thereby minimizing localized plastic deformation. This enhances the efficiency of load transfer within the composite, as the load is transmitted from the filler to the matrix, even in cases of weak interfacial bonding. At higher concentrations, MoS_2 particles establish strong interfacial bonds with the PA-6,6 matrix, further improving load transfer efficiency and contributing to the material's enhanced hardness. The findings of the present study align well with existing literature, confirming the observed improvement in hardness [39, 54, 89]. Moreover, the hardness influences the material's surface deformation and abrasive wear phenomenon. A higher hardness generally correlates with lower wear rates, as it reduces material removal during sliding contact [90, 91].

4.6.2 | Tensile Behavior

The quasi-static tensile test was performed on the developed PM composites. The study investigates the influence of the MoS_2 filler contents in various concentrations on mechanical properties. The study of the mechanical properties of composite systems is essential and possesses great significance in understanding the materials' behavior under tension and its fracture phenomenon, which further helps in estimating the effect of filler contents in filler–matrix adhesion [75] and its distribution over the matrix region [15, 16, 92]. The present findings are found to be significantly higher than the previous studies [93–95].

Figure 7a,b delineates the tensile properties of the PM composite as a function of the MoS_2 filler reinforcement in various concentrations. A significant increasing trend in the tensile modulus characteristics can be seen in Figure 7a. A systematic reinforcement of MoS_2 filler contents at different wt.% enhanced the tensile modulus significantly. The maximum modulus was found to be 10 wt.% of MoS_2 filler addition. An enhancement in tensile modulus was found to be ~48%, ~58%, ~63%, and ~94% than pure PA-6,6 at 1 wt.%, 3 wt.%, 5 wt.%, and 10 wt.% of filler addition, respectively. Similarly, an increasing trend in tensile strength characteristics up to 5 wt.% of MoS_2 filler addition can be observed in Figure 7b, and at 10 wt.% of filler loading, decreased tensile strength characteristics were observed. The tensile strength characteristics were found to be increased by ~13%, ~15.5%, and ~21% compared to pure PA-6,6 at 1 wt.%, 3 wt.%, and 5 wt.% of MoS_2 filler loadings, respectively. The higher wt.% (say 10 wt.%) of MoS_2 reinforcement decreases tensile strength by ~4% compared to pure PA-6,6. This improved tensile behavior of the PM composite indicates the introduction of a stiffening effect due to MoS_2 reinforcement. The incorporation of MoS_2 enhances the interaction of MoS_2 particles with the PA-66 matrix due to its self-lubricating property. This is due to the presence of solid interactions like H-bonding between the MoS_2 particles and the PA-6,6 matrix [96]. The polymer attached to the 2-dimensional surface of MoS_2, due to H-bonding, generates the nano-confined region, which effectively improves the volume fraction of the composite and facilitates effective stress transfer from fillers to the matrix phase [96, 97].

Additionally, the enhancement in tensile strength characteristics (up to 5 wt.%) may be due to the good compatibility of MoS_2 fillers with the PA-6,6 matrix, but apart from imparting a positive effect on tensile strength characteristics. However, the excessive loading of MoS_2 (at 10 wt.%) exhibited an opposite effect, as shown in Figure 8b. The higher loading of MoS_2 may lead to the formation of nanochannels and defects due to the uneven and inefficient wrapping of filler contents with

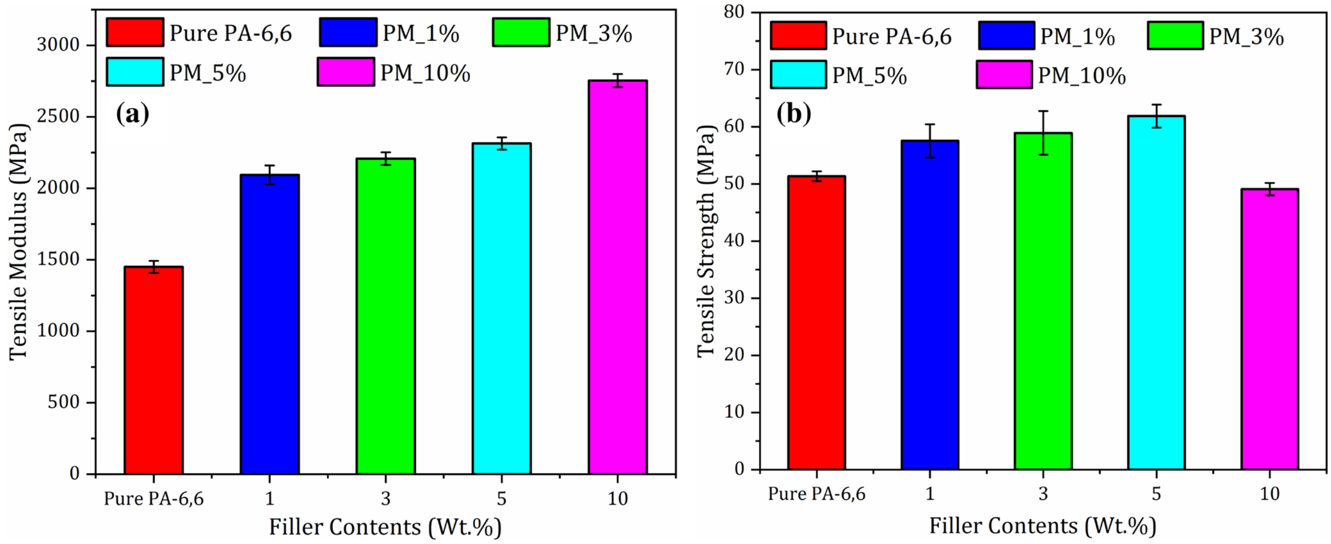

FIGURE 7 | Mechanical properties of PA-6,6/MoS_2 illustrating the (a) tensile modulus and (b) tensile strength characteristics. [Color figure can be viewed at wileyonlinelibrary.com]

the matrix phase. Also, the higher wt.% of filler addition may increase the possibility of filler aggregation over the matrix region [98]. This leads to a sudden decrease in tensile strength [99]. Additionally, the lamellar structure of MoS_2 develops the shear slip under stress development in the composite [100]. Therefore, the study concludes that 5 wt.% of MoS_2 filler contents exhibit an optimum tensile property. The current study agrees well with the previous literature reported by Mittal et al. [101] and Puértolas et al. [102] and Kumar et al. [15, 16]. Furthermore, the higher strength is an essential factor in determining the load-bearing capacity of the composites under tribological conditions and justifies the enhanced wear resistance characteristics by preventing severe deformation or fatigue failure [70, 103–105]. Therefore, to establish these correlations, the wear behavior of the PM composites in relation to the enhanced mechanical properties has been discussed in the subsequent section to better comprehend the wear mechanisms.

4.6.3 | Fractured Surface Morphology

The FESEM images of the fractured surface were used to better comprehend the fracture mechanisms of the developed PM composites with MoS_2 reinforcement at different wt.%. Figure 8 shows the photomicrographs of the tensile fractured surface morphology of the pure PA-6,6 and its PM composites with different concentrations of MoS_2 reinforcements. Figure 8a,b demonstrates the smooth, flat morphology with ridges and microcracks. The material pull out exhibits a borrow-pit structure (see Figure 8b), showing a fracture dimple, suggesting a ductile fracture of the pure PA-6,6 matrix. The developed microcracks (indicated in small red arrows) during fracture may possibly cause the ductile fracture. Further, the MoS_2-filled PM composite demonstrates a smooth river and tongue-like pattern (see Figure 8c–f). Figure 8c shows a flower-like fractured surface with associated cracks demonstrating a fractured gully and a tearing-shaped pattern. Such a tearing pattern shows the extruded fracture of the material, which demonstrates that at low wt.% (i.e., 1 wt.%) of filler addition does not affect the ductility effectively, and the material underwent ductile fracture. Figure 8d–f shows similar material pull-out and developed cracks and microcracks. Figure 8e demonstrates a fiber-like cracking with extruded fractured material exhibiting a ravines-like structure along with the developed cracks and microcracks showing gully-like morphology. Smooth surface morphologies demonstrate the good dispersion of MoS_2 particles up to 5 wt.% of filler loadings in the PM composites. This indicates the good compatibility of filler and matrix material, which further leads to crack deflection and causes the absorption of tensile fracture energy that further hinders crack propagation. This, in turn, increases the tensile strength characteristics of the PM composite.

Additionally, some possible aggregation (indicated by the blue circle) of the filler materials can also be seen in Figure 8f, demonstrating the fact that higher wt.% (i.e., 10 wt.%) of the filler addition may possibly cause the filler agglomerations and some aggregation sites can also be observed in Figure 8f. At 10 wt.% of MoS_2 filler reinforcement, the squamous-like morphology with a relatively rough surface and detached extruded material was identified, attributing to enhanced brittleness characteristics of the composite due to higher contents of MoS_2 filler particles. the higher concentration of MoS_2 particles causes interlaminar slippage among the stacked layers of MoS_2, which further promotes the discontinuity of the matrix phase and leads to the formation of microcracks (indicated by the blue circle in Figure 8f over the entire region), which potentially behaves as the stress concentration zones, that further lead to the crack initiation and propagation in the materials and premature failure of the material occurs. Furthermore, the agglomeration (shown by the blue circle in Figure 8f) indicates an inhomogeneous dispersion and poor filler–matrix interfacial adhesion. This indicates the poor retention or filler binding capability of PA-6,6, demonstrating

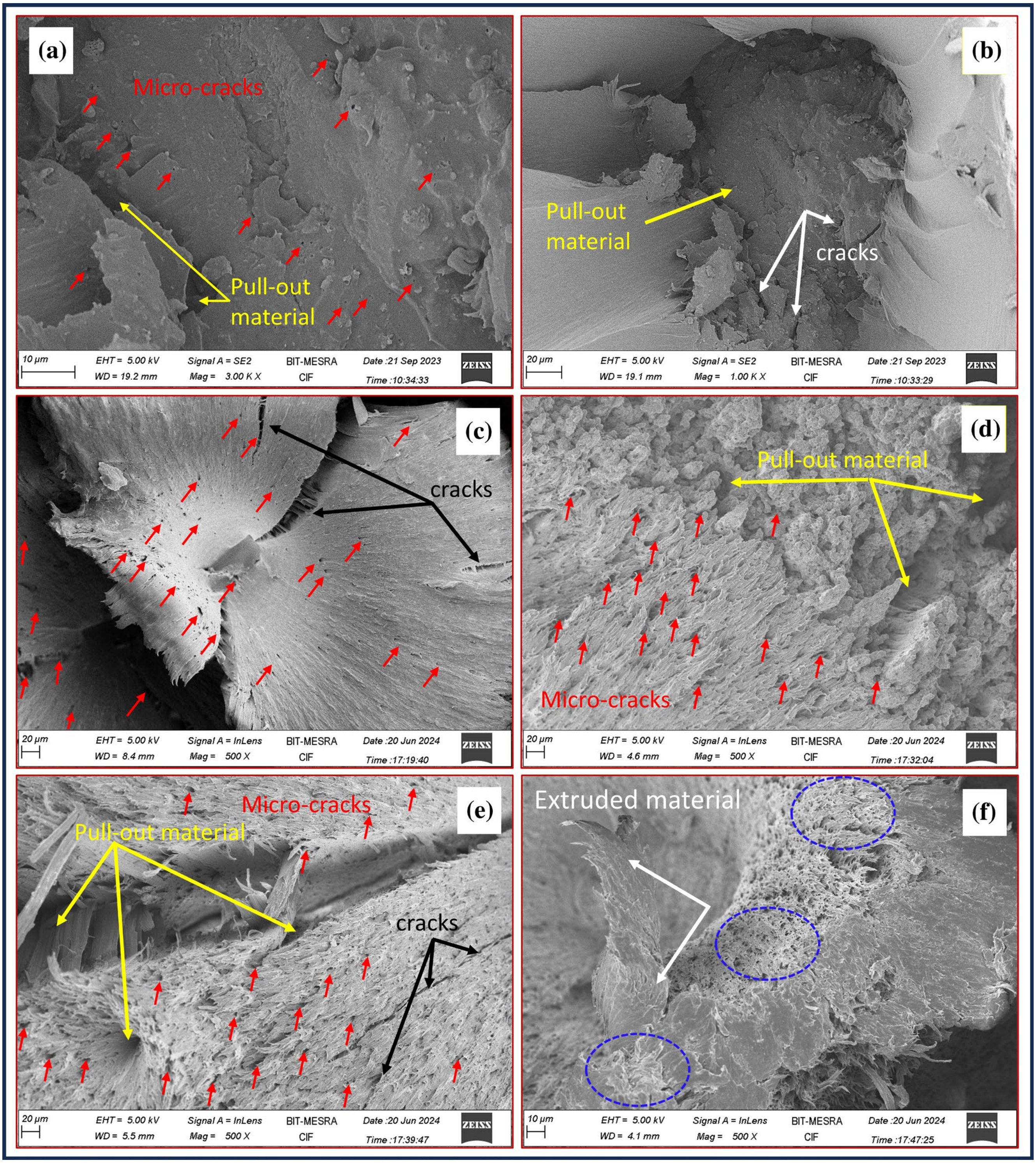

FIGURE 8 | FESEM micrographs demonstrating the fractured surface morphologies of (a) and (b) pure PA-6,6, (c) 1 wt.%PA-6,6/MoS_2 nanocomposite, (d) 3 wt.%PA-6,6/MoS_2 nanocomposite (e) 5 wt.%PA-6,6/MoS_2 nanocomposite, and (f) 10 wt.%PA-6,6/MoS_2 nanocomposite. [Color figure can be viewed at wileyonlinelibrary.com]

reduced load-carrying capability. Therefore, the developed microcracks, wrinkles, and aggregation of the filler contents at higher concentrations appearing at the fractured surface of the PM composite may cause premature failure, which leads to reduced tensile strength.

4.6.4 | Impact Behavior

While the influence of the filler particles on tensile strength is relatively limited, their influence on impact strength is significant [3]. Figure 9 delineates the impact test results for pure

PA-6,6 and pure PA-6,6-based composites for varying wt. fraction of MoS_2 filler reinforcements. Figure 9 records that the pure PA-6,6 matrix experiences a maximum impact strength of about 50.147 ± 1.463 J/m [16]. Furthermore, a continuous downtrend was exhibited by the impact test results. The reduced impact strength of the PM composite is due to the synergistic effect of the MoS_2 filler addition in varying wt. fraction (wt.%). The ductile nature of the pure PA-6,6 matrix can demonstrate the superior impact strength of PA-6,6. However, with the reinforcement of MoS_2 filler particles, the impact strength was reduced from 9.8% to 24.6% compared to pure PA-6,6. The maximum reduction in impact strength of 24.6% was observed at 10 wt.% of MoS_2 filler addition. At 3 to 5 wt.% of filler addition, an almost linear trend in impact strength was observed. Such reduced impact strength was found to be consistent with prior research [106]. Also, the gradual increase in filler loadings gradually imparts the brittle characteristics in the composite system, resulting in a reduced impact strength. Furthermore, The higher filler addition usually reduces the impact strength and may be attributed to the agglomeration of filler particles [3, 107].

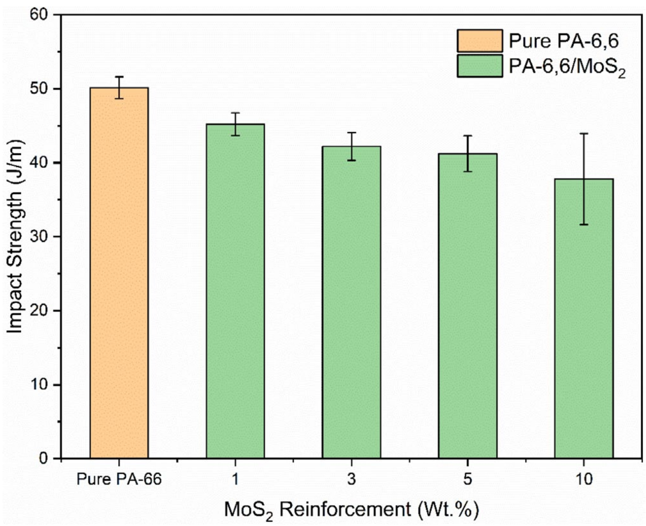

FIGURE 9 | Impact strength of PA-6,6 and MoS_2 reinforced PA-6,6 composite. [Color figure can be viewed at wileyonlinelibrary.com]

Excess filler content in the composite system leads to poor particle-to-particle interaction, inadequate particle–matrix adhesion, and poor dispersion of filler particles within the matrix phase. This can result in the development of induced voids and uneven mixing of fillers and the matrix [108, 109]. Additionally, the accumulated filler content within the matrix increases the possibility of microcrack formation, which acts as internal notches. These notches can cause sudden fractures under impact loading, ultimately reducing the impact strength [43].

4.7 | Tribological Properties

4.7.1 | Friction and Wear Behavior

The wear and mechanical behavior of polymeric composite materials are greatly influenced by parameters such as shape and size, the types of filler reinforcement, chemical treatments, and modifications [39]. Thus, this study aims to utilize solid lubricant-based filler materials to enhance the capability of stress transfer and minimize wear rates. The present study was carried out to assess the frictional coefficient and specific wear rates under abrading conditions. The tribological performance of the pure PA-6,6 and its PM composites was analyzed under different loads and abrading frequencies.

Figure 10a,b display the change in CoF as a function of different wt.% of filler material for all loads and abrading frequencies for pure PA-6,6 and its PM composites. As seen in Figure 10a,b, a continuous decreasing trend in CoF behavior was identified, and the friction coefficient was reduced with increasing concentration of MoS_2 filler contents. The CoF of the PM composites was found to be reduced up to 5 wt.% of the MoS_2 filler addition. Subsequently, an increasing effect was identified for higher filler reinforcement of 10 wt.% for the applied normal loads from 10 to 50 N under both frequencies.

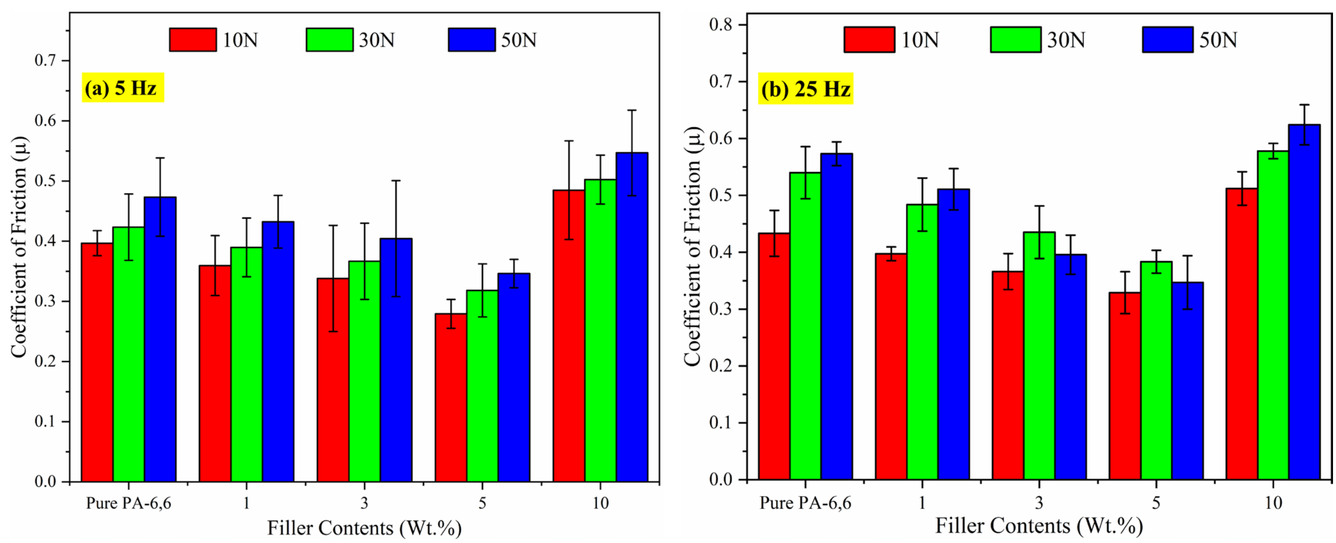

FIGURE 10 | Coefficient of friction versus filler contents under (a) 5 Hz frequency and (b) 25 Hz frequency. [Color figure can be viewed at wileyonlinelibrary.com]

An ideal situation of a marked tribological behavior of a material is associated with lower friction values with minimum frictional fluctuations. Therefore, the developed PM composites exhibited superior results in the present case compared to pure PA-6,6. The prevailing explanation of CoF, μ, is given by Equation 3 [39, 110], where the applied normal load (N) and constants K and m are considered, with the value m varying from 2/3–1, which further depends on the interaction between the plastic and elastic deformations. In the present study, compared to PA-6,6 and its composite, the systematic reinforcement of solid lubricant MoS_2 filler at different wt.% exhibited a significant reduction in CoF behavior. The filler reinforcement at 5 wt.% showed a marked reduction in CoF by ~66%, ~57%, and ~47% for 10, 30, and 50 N normal load, respectively, under 5 Hz of abrading frequency compared to pure PA-6,6.

$$\mu = K \times N^{(m-1)} \tag{3}$$

Similarly, at the filler reinforcement of 5 wt.% and under 25 Hz abrading frequency, the reduction in CoF was observed to be ~24%, ~25%, and ~16% for 10, 30, and 50 N loads, respectively, as shown in Figure 10b. Thus, the present study reveals that adding solid lubricant MoS_2 solid lubricant material as filler particles to PA-6,6 resulted in more consistent friction coefficients, as indicated by reduced frictional fluctuations and lower standard deviation. The CoF characteristics for PM composites decreased more significantly than for pure PA-6,6, indicating that a 5 wt.% of MoS_2 filler addition was the optimum and more effective filler concentration in reducing friction under increasing load and abrading frequency.

Furthermore, Figure 11a,b indicates the influence of different loads with various filler concentrations under all frequencies and applied loads. The PM composites exhibited a significantly improved wear behavior compared to the pure PA-6,6. The MoS_2 filler reinforcement at 3 to 5 wt.% indicated an appreciable wear resistance property. At 3 wt.% of MoS_2 reinforcement and under 5 Hz frequency, the decrease in wear rate was found to be ~60%, ~51%, and ~59% for all loads, respectively. Whereas, under similar conditions and at 5 wt.% of MoS_2 reinforcement, the W_{SR} characteristics were reduced by ~64%, ~67%, and ~73%. Similarly, under 25 Hz abrading frequency and all load conditions, a higher wear behavior of the material was observed compared to the 5 Hz abrading frequency level. The 3 wt.%–5 wt.% of MoS_2 addition indicated good wear resistance behavior. At 3 wt.%, the wear rate was reduced by ~65%, ~59%, and ~56% for all loads, respectively. Also, at 5 wt.% of MoS_2 reinforcement, the wear rate was found to be decreased by ~72%, ~78%, and ~79% for all loads, respectively. Moreover, a higher wt.% (i.e., 10 wt.%) of filler addition indicated an opposite result with an increasing trend in W_{SR} characteristics. The wear behavior of solid lubricant MoS_2 reinforced polymeric composite seems appreciable due to its wear resistance property and efficient load transfer capability. However, the matrix itself plays a vital role. It entraps the filler particles strongly by proper mixing and strong bonding with the extruded melt blend. Moreover, the efficacy of these fillers depends upon the types of filler/fiber reinforcement, the volume of filler/fiber reinforcement, aspect ratio, particle size, matrix properties, and their filler–matrix interaction or filler–matrix bonding strength as well as uniform dispersion over the entire region of the matrix phase [39, 111]. One possible mechanism is the transfer film formation during the continuous sliding action and abrasion, which reduces the degree of abrasiveness considerably.

Furthermore, it is worth noting that for sliding wear of polymers against metallic counterface surfaces, the friction components due to the adhesive mechanism equal the product of the real contact area and the shear stress of the softer material [112, 113]. The polymer experiences frictional heating with an increase in the normal load, leading to an increased contact area due to the high contact pressure at the interface of the polymer surface and the counterpart surface. This results in two contrary effects on the friction coefficient. Firstly, the CoF decreases due to a decrease in the shear strength of the softened material. Secondly, the elastic modulus of the polymers decreases at elevated temperatures, which further results in an increased real contact area and restricts the decrease in the

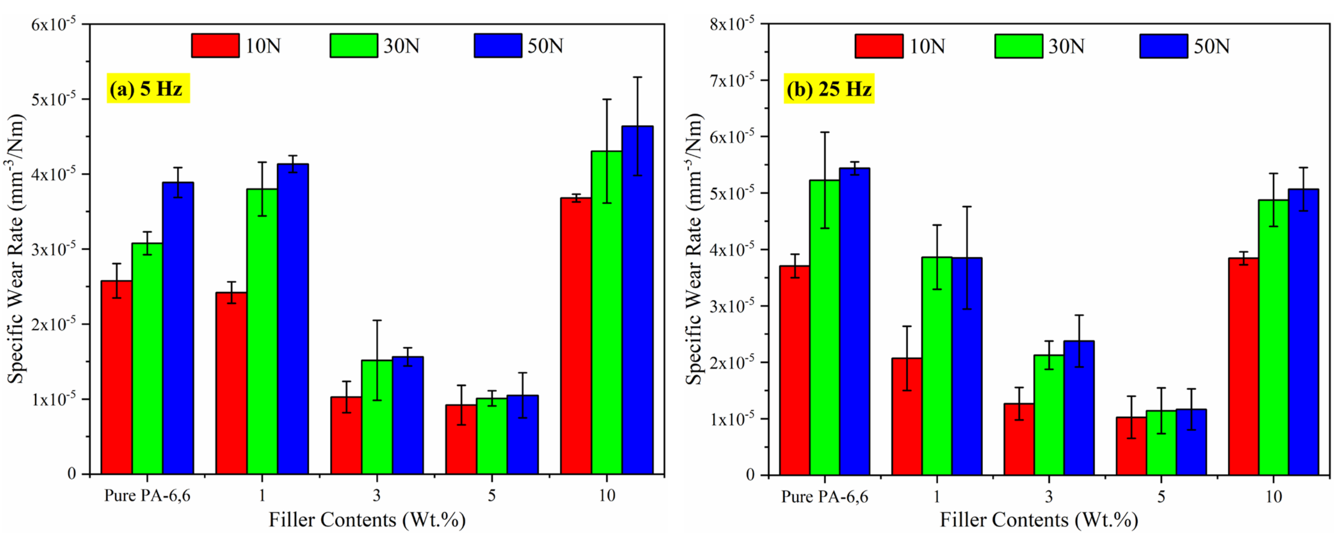

FIGURE 11 | Specific wear rate as a function of the filler contents (wt.%) against the different applied loads for pure PA-6,6 and PM composites under (a) 5 Hz frequency and (b) 25 Hz frequency. [Color figure can be viewed at wileyonlinelibrary.com]

friction coefficient [114]. In the present case, the friction coefficient was increased with an increase in normal loads from 10 to 50 N due to frictional heat being dominant. Thus, the reason for the continual increase in CoF with the increase in normal loads for all wt.% of MoS_2-filled PM composites was identical. Additionally, with an increase in abrading frequency from 5 to 25 Hz, the CoF and W_{SR} characteristics were found to be significantly increased due to high levels of frictional heat generation during the abrasion mechanism with increasing normal loads [70]. The high frictional heat at the contact interface softened the substrate plate, causing the material to extrude during the sliding action due to the imparted high shear force [39, 69, 70].

Moreover, the observed decrease in W_{SR} characteristics with increasing applied load is found to be consistent and possesses good agreement with the previous findings [44, 45, 115, 116]. Also, it is a well-known fact that most of the thermoplastic materials exhibited a higher wear rate at an early stage of abrasion. At the beginning of abrasion, the high stress generated due to the high load is sufficient to induce the failure of the polymer's surface layer, leading to a high material removal rate. This also diversely affects the surface with debonding and more debris generation due to increased applied load. As a result, a reduced wear rate is observed due to the entrapment of debris at the contact of the tribo pair [47, 48].

4.7.2 | Worn Surface Morphology

The worn surface morphologies of the PA-6,6 and its PM composites were evaluated using the SEM images, which are illustrated in Figures 12, 13, and 14 for visual reference and examination. The arrow indicates the sliding direction. The worn-out surface of the PA-6,6 matrix under all abrading frequencies and all applied loads can be seen in Figure 12. The worn surface exhibits a rough texture, with material pull-out and wider grooves. Additionally, cracks on the worn surface were also observed. The SEM observation reveals that during the high heat generation and increased load (10–50 N), the melt polymer accumulated on the worn surface, and the scratches, material peeling, and microcracks became severe.

Figure 13 shows minimal surface damage and improved wear behavior of the PM composites under low abrading frequency. The SEM micrographs presented in Figure 13 with 5 and 10 wt.% of MoS_2 filler reinforcement indicate mild wear at low load (30 N). As the load increases, the improved wear behavior in worn surfaces with a relatively smoother surface than pure PA-6,6 can be clearly observed. At 5 and 10 wt.% of MoS_2 reinforcement, as shown in Figure 13, a relatively smoother and uniform surface with minimal surface damage and the presence of little debris with fine scratches can be seen. A foreign particle can also be observed that has been left behind during the continuous sliding

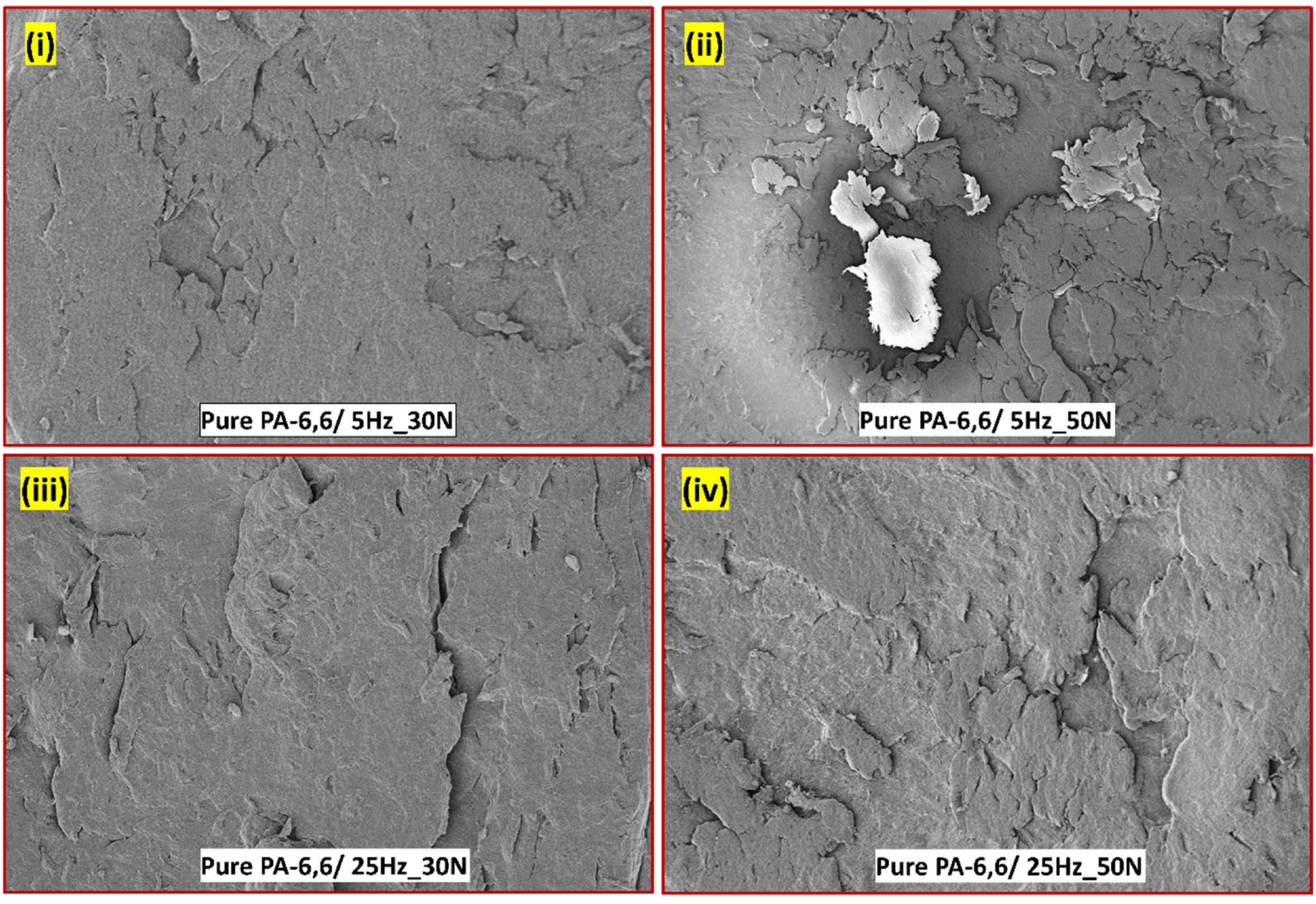

FIGURE 12 | The SEM micrographs indicate the worn-out surfaces of pure PA-6,6 subjected to low abrading frequency (i and ii) and high abrading frequency (iii and iv) under all loads. [Color figure can be viewed at wileyonlinelibrary.com]

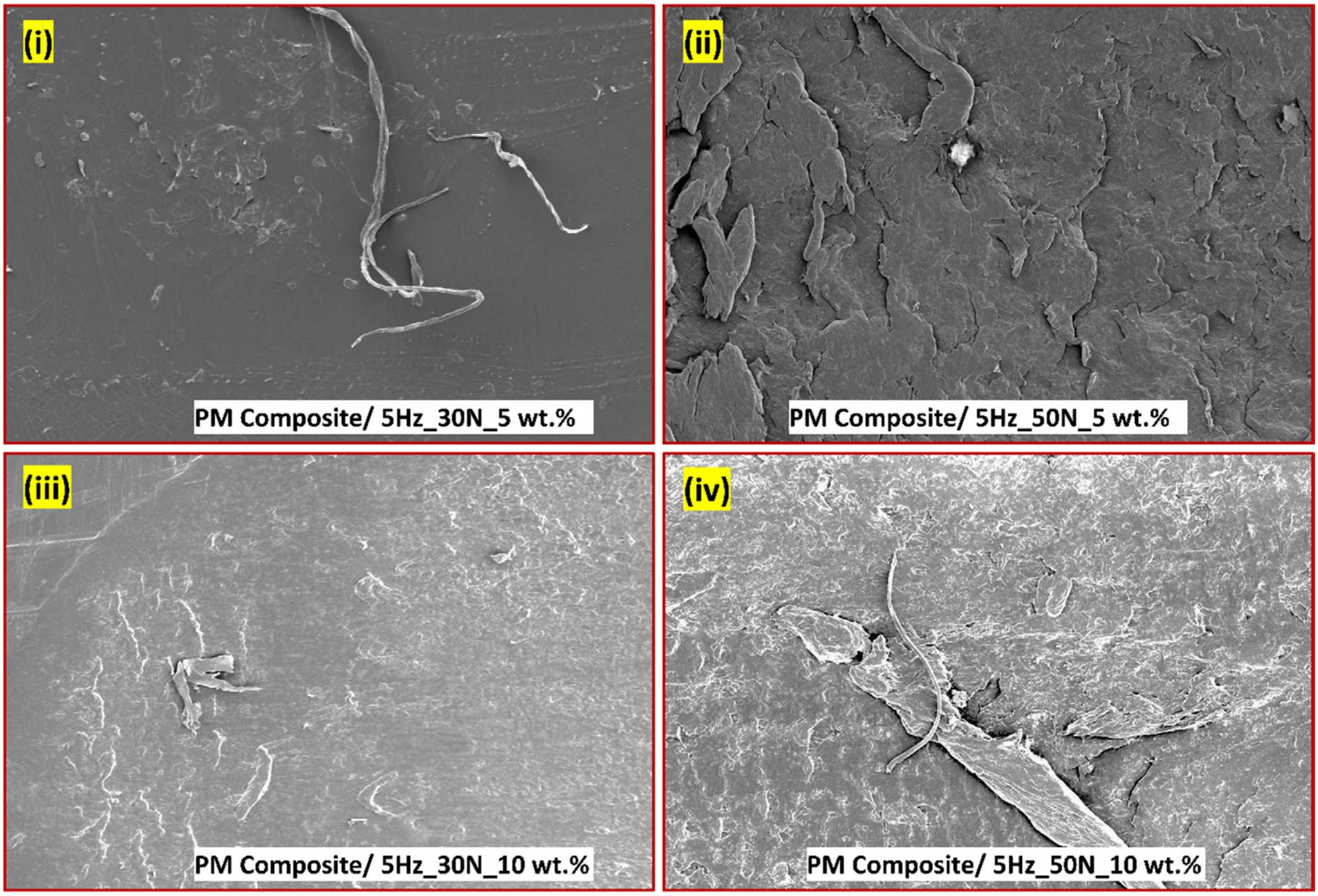

FIGURE 13 | The SEM micrographs indicate the worn-out surfaces of PM composites subjected to low abrading frequency (i–iv) under all loads for 5 wt.% and 10 wt.%. [Color figure can be viewed at wileyonlinelibrary.com]

action and transfer layer formation. Additionally, at a high load (50 N), the plastically deformed material can be seen, which accumulated on the worn surface, indicating a severe adhesive wear mechanism. This may be due to the fact that at high loads, the interfacial bonding strength becomes weaker and severe plastic deformation occurs due to high frictional heat generation, which further softens the material; the abrasion of the melt blend accumulates over the worn surface and results in a high wear rate at increased load. Furthermore, elastic and plastic deformation signifies the main cause of wear behavior under the inclusion of nanoparticles [3].

The wear mechanisms of pure PA-6,6 and its PM composites at high abrading frequency were complex due to the extreme load and frequency conditions involved. The key wear mechanism involved was adhesive wear, which occurred due to the sliding contact of the PA-6,6 composites and the steel ball counterface surface tribo pair, leading to the adhering material to the surface of each other due to high pressure and temperature at high abrading frequency. Figure 14 resulted in material transfer from one surface to another, leading to the formation of wear debris and surface damage. Due to high load and increased abrading frequency, the worn surface experiences several cracks, microcracks, and increased contact and temperature, which further leads to material removal due to the presence of hard particles. This severe material removal mostly occurs due to high filler reinforcement, the plowing action, cutting/microcutting, as well as high shearing, resulting in surface erosion [39]. During the wear process, void formation was initiated, lumps of the melt blend formed, and relatively larger wear debris was eventually generated. Additionally, thermal softening and melting were observed at high load and abrading frequency with higher wt.% of filler addition, which can be clearly observed in Figure 14. Finally, surface deformation occurs due to high abrading frequency, and the included plastic deformation leads to a change in mechanical properties such as hardness, elastic modulus, and, ultimately affecting the wear mechanism.

Additionally, the excessive reinforcement may hinder the desired property due to improper wetting of the filler particles by the matrix material, resulting in the accumulation of filler particles at various locations over the matrix region and poor filler–matrix adhesion and filler–matrix bonding [54, 69, 101]. Under high mechanical and thermal stresses during the sliding or rubbing action in the tribo pair, the poorly bonded filler particles may peel off or pull out from the surface, leading to an increase in W_{SR}. Also, due to high frictional heat generation, the softened matrix may undergo microcutting and pulverization, further enhancing the W_{SR} characteristics [117]. Similar studies have also been reported in the previous literature [118, 119]. The enhanced wear performance and the wear resistance property are attributed to the restraining effect due to the self-lubricating property in direct contact between the soft polymer and hard steel ball counterface [120, 121].

4.7.3 | Formation of Transfer Film

The transfer film of PM composites formed on the counterface surfaces is shown in Figure 15a,b. The SEM images delineated in Figure 15a revealed that PM composites produced a fairly thick and lumpy film, with deposited composite materials as a transfer film under high abrading frequency. This indicates that the PA-6,6 gets transferred to the steel ball counterface surface during sliding action due to MoS_2 reinforcement. Moreover, the softening of the PM composite under high abrading frequency and high load conditions contributed to forming a uniformly thin layer of transfer film with full

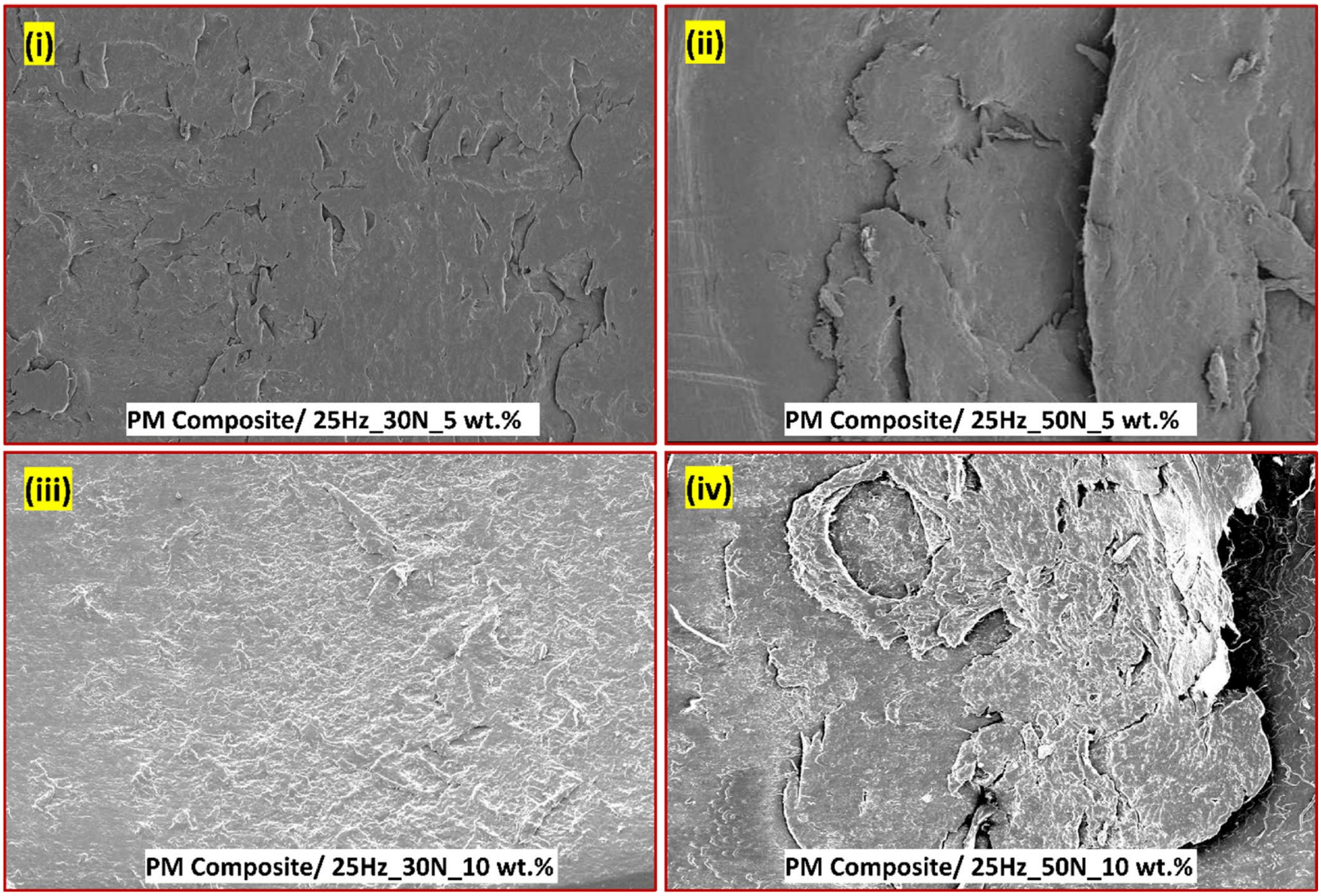

FIGURE 14 | The SEM micrographs indicate the worn-out surfaces of the PM composites subjected to high abrading frequency (i–iv)under all loads for 5 wt.% and 10 wt.%. [Color figure can be viewed at wileyonlinelibrary.com]

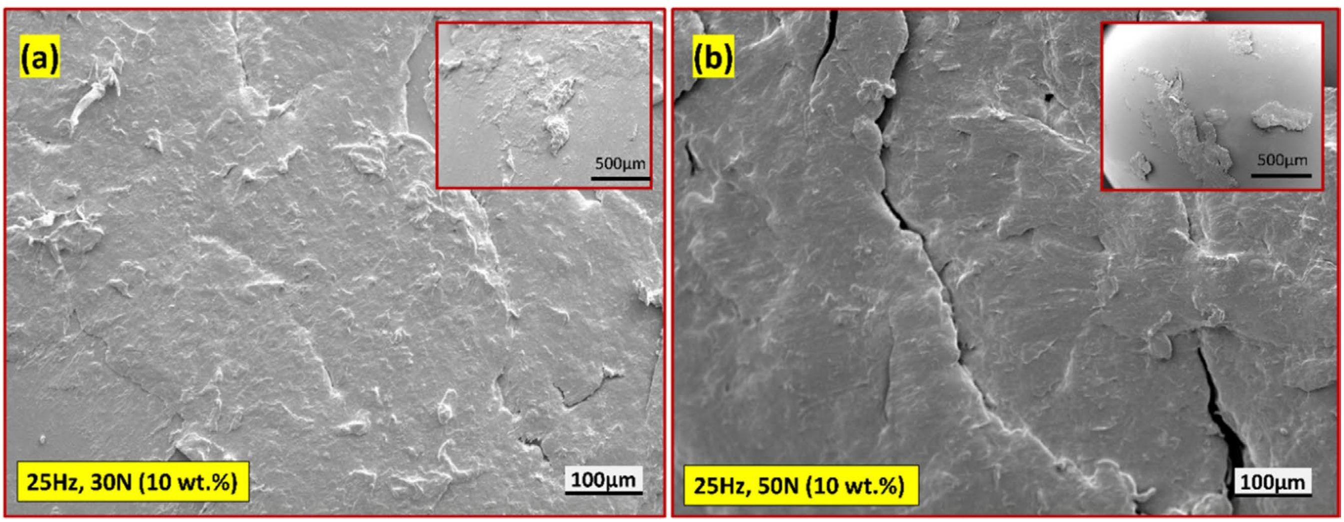

FIGURE 15 | SEM images showing the transfer layer film formation on the steel ball counterface surface. [Color figure can be viewed at wileyonlinelibrary.com]

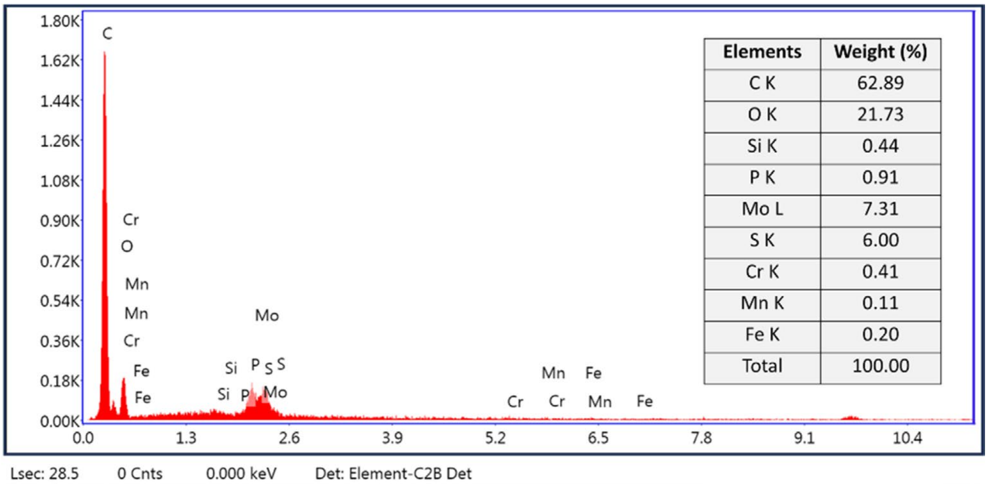

FIGURE 16 | EDS of the formed transfer layer film on the steel ball counterface surface. [Color figure can be viewed at wileyonlinelibrary.com]

coverage (see Figure 15a). This may be due to the plowing action during the continuous dry sliding contact under high abrading frequency. Similarly, in Figure 15b, a thick, nonuniform, and noncontinuous transfer layer was observed. This may be due to the presence of MoS_2 at higher concentrations that are subjected to high abrading frequency and high load. Therefore, the high abrading frequency at a higher load, along with the higher reinforcement of filler particles, significantly influences the transfer layer formation. It was noteworthy that the transfer film was also observed to be detached from the counterface surface during continuous dry sliding and was left behind and accumulated as a melt blend over the worn surface as various slices, as discussed previously (see Figures 13 and 14), leading to an adhesive wear mechanism.

Furthermore, to verify the presence of MoS_2 within the transfer film, the EDS spectrum analysis, as shown in Figure 16, was carried out. The peaks corresponding to the C, O, Si, P, Mo, S, Cr, Mn, and Fe confirm the presence of MoS_2 reinforcement in PM composites and the steel ball compositional elements that approve the associated sliding action over the worn surface. This observation indicated that MoS_2 filler contents are involved in chemical reactions during the sliding action of the tribo pair. The previously reported research has also suggested that a proportion of the MoS_2 filler contents could be oxidized to form MoO_3 [115, 122, 123] during the sliding process. The oxidation of MoS_2 leads to the formation of FeS, $FeSO_4$, and $Fe_2(SO_4)_3$ due to the decomposition of pure Sulfur from MoS_2 [115]. In previous research, several authors have reported the influence of MoO_3 on the friction and wear characteristics of the polymer composite. Also, the MoO_3-nylon transfers easily to the counterface, but the adhesive strength of the transfer film is relatively weak, making it rub off easily, which results in a high wear rate [124, 125]. A similar phenomenon was identified in the current findings.

4.8 | Water Contact Angle (WCA) Measurement

The surface wettability of the pure PA-6,6 and its PM composites at different wt.% of MoS_2 reinforcement were evaluated by measuring WCA. This is one of the essential and effective parameters in terms of the tribological behavior and adhesion properties of the filler contents at the contact interface of the material [126]. The water shapes of droplets and the WCA plot are delineated in Figure 17 for pure PA-6,6 and the prepared PM composites [54, 127]. Figure 17 exhibited an increasing trend in the WCA behavior of the PM composites as the MoS_2 concentration increased from 1 wt.% to 10 wt.%, with a decreasing trend in WCA after adding MoS_2 filler contents at different concentrations. The pure PA-6,6 polymer exhibited a relatively lower contact angle with an average value of 72.5°, indicating a hydrophilic nature of the pure PA-6,6 surface. As the MoS_2 fillers were added in different concentrations, the WCA values significantly increased, suggesting a transition to a more hydrophobic surface behavior of the PM composites. Furthermore, the increasing WCA values correspond to a decreasing surface free energy (SFE) of the composite. This may be attributed to the hydrophobic nature of MoS_2 when reinforced in higher wt.%, which further reduces the surface free energy by minimizing the interaction between the composite surface and water contact. MoS_2 particles create a rougher texture and a less adhesive surface, thereby lowering the overall wettability of the material. The reduction in surface

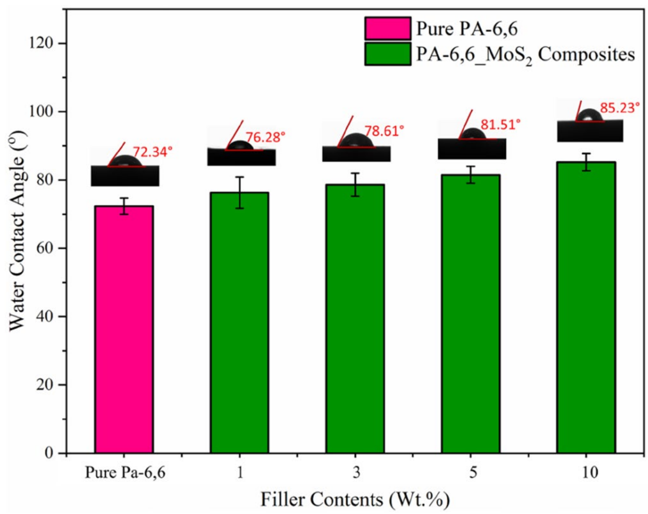

FIGURE 17 | Variation in water contact angle (WCA) for PA-6,6 and its PM composites. [Color figure can be viewed at wileyonlinelibrary.com]

energy with increasing MoS_2 content is a direct indicator of enhanced hydrophobicity [69]. The observed results in the present work were found to be more consistent with the previous work [69, 128, 129]. Additionally, The decreasing surface energy with increasing MoS_2 reinforcing concentration correlates with improved tribological performance, making the developed PM composites suitable for applications requiring low friction and high wear resistance.

Furthermore, increased hydrophobicity with lower surface free energies directly affects PM composites' friction and wear behavior. Lower SFE decreases adhesion between the composite surface and the steel ball counterface surface, reducing the frictional resistance during the sliding contact. The reduced interaction and lower surface adhesion also contribute to decreased material transfer and wear, leading to an improved wear resistance characteristic. Furthermore, the presence of solid lubricant MoS_2 enhances the self-lubricating properties of the composite. At higher MoS_2 concentrations, a lubricating layer forms at the interface during sliding, which acts as a protective barrier, preventing direct contact between the surfaces and reducing both friction and wear.

5 | Conclusion

This study investigated the physical, structural, thermal, mechanical, and tribological properties of injection-molded PA-6,6 and its MoS_2-reinforced PM composites. The research highlights the significant influence of MoS_2 reinforcement on enhancing the material's performance across multiple aspects. The key outcomes are summarized as follows:

The MoS_2 reinforcement in PA-6,6 significantly enhanced the thermal stability of the composites. At 10 wt.% filler loading, the thermal degradation temperature increased by approximately 13%, indicating improved resistance to high-temperature environments, as further confirmed by XRD and DSC analysis.

The addition of MoS_2 reinforcement resulted in notable increases in density and Vickers hardness, with the highest increments of approximately 9% and 60%, respectively, observed at higher filler concentrations. The tensile strength improved by 12%, 15%, and 23% with 1, 3, and 5 wt.% MoS_2, respectively, demonstrating the reinforcing effect of MoS_2. However, the impact strength showed the opposite effect with increasing filler content. Hence, a careful consideration of impact strength along with mechanical strength is necessary.

The PM composites exhibited superior tribological properties, particularly at 5 wt.% MoS_2. The coefficient of friction (CoF) decreased by up to 79%, and the wear rate reduced by up to 73% under varying abrading frequencies and normal loads, emphasizing the role of MoS_2 in improving wear resistance and self-lubrication. The worn surfaces of the composites displayed smoother morphologies compared to pure PA-6,6, indicating reduced wear severity and enhanced durability under different operating conditions. In summary, incorporating MoS_2 into the PA-6,6 matrix significantly enhances its structural, thermal, mechanical, and tribological performance.

The enhanced wear performance of the PM composites makes them suitable candidates for applications where low friction and high durability under dynamic loading conditions are required. Based on the present findings, the PM composites could be potentially used in automotive components (gears, bushings, bearing cages), industrial machinery (conveyor rollers, seals, sliding elements), aerospace components, biomedical devices, and consumer electronics. The present findings may help engineers optimize the material selection for enhanced wear resistance properties in dynamic conditions. Future investigations could focus on exploring the electrical and thermal conductivity of these composites to expand their application scope into electronic and thermal management systems.

Author Contributions

Ranjan Kumar: data curation (lead), formal analysis (lead), investigation (lead), writing – original draft (lead). **Sujeet Kumar Mishra:** supervision (supporting), validation (supporting). **Sudeepan Jayapalan:** conceptualization (lead), methodology (lead), supervision (supporting), writing – review and editing (lead).

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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